CLOUD COMPUTING LAB - 7

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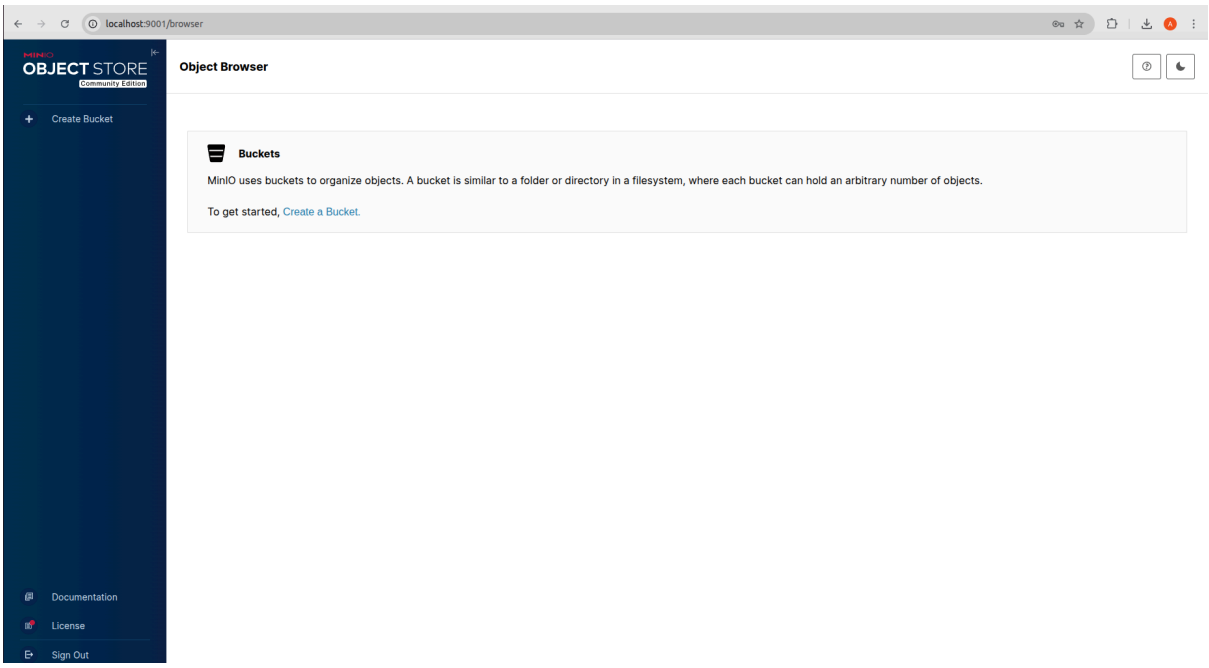
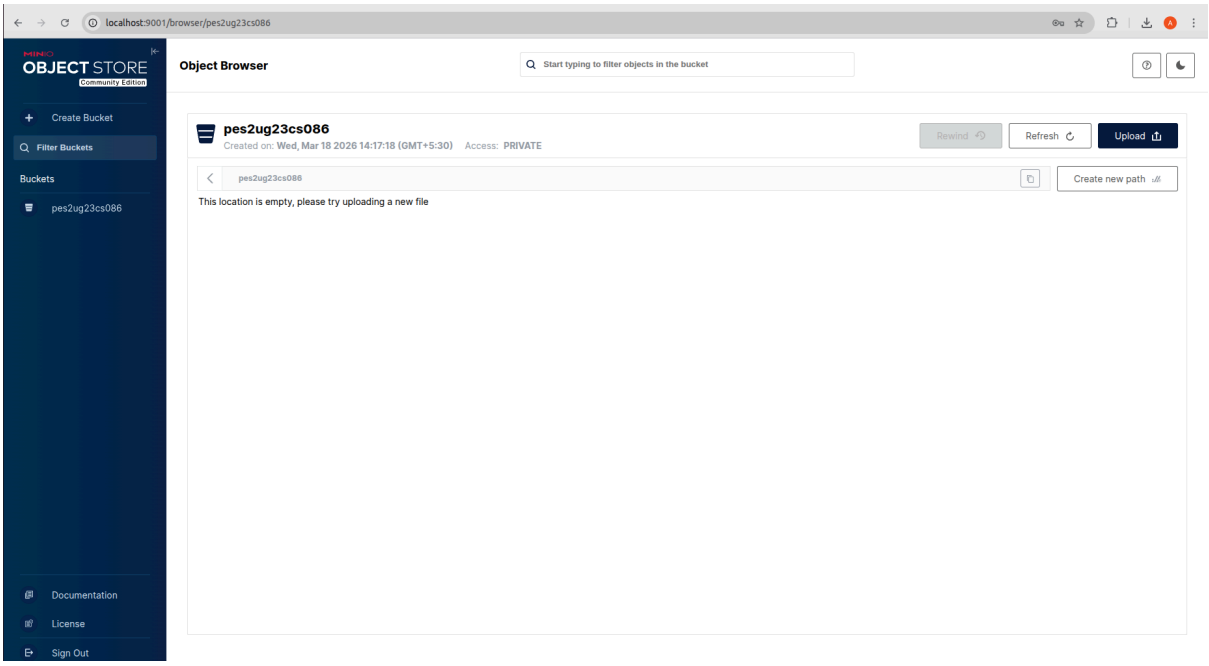
SRN : PES2UG23CS086
SECTION : B

SS1

SS2

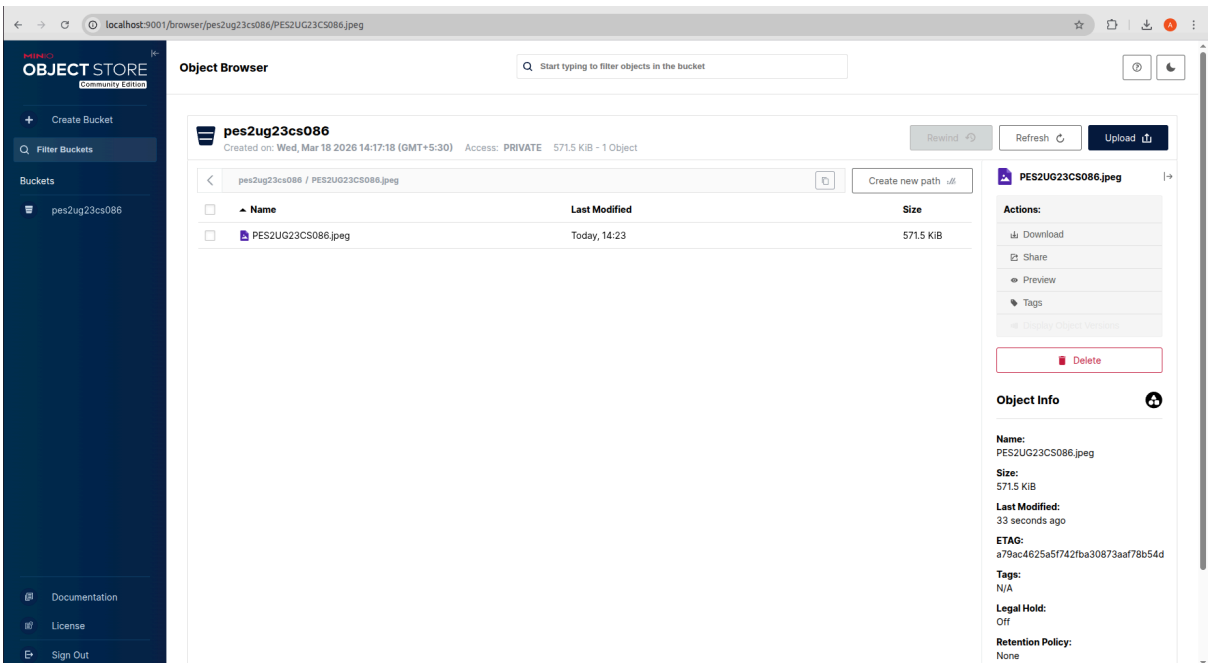


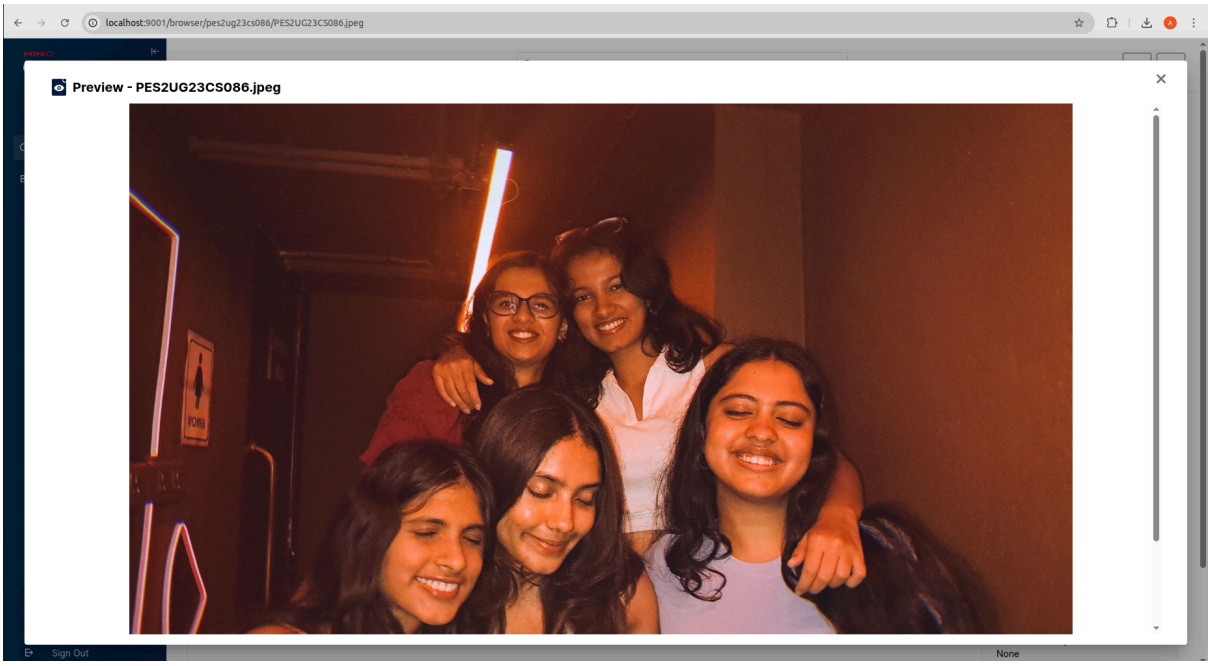
SS3



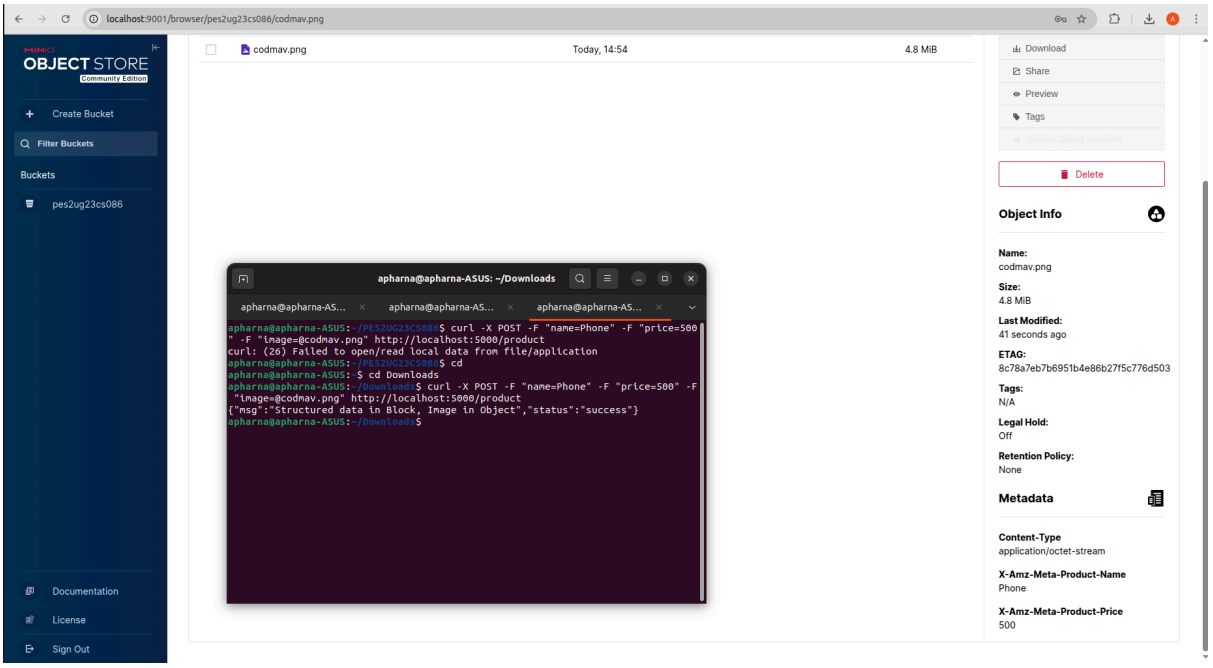
SS4 (experiment 1)

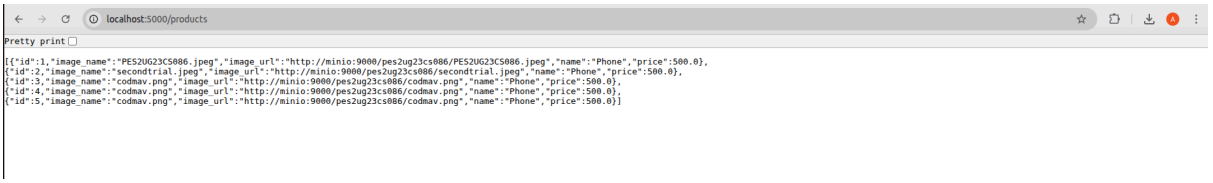
Block storage (Ceph) provides direct file system access and is suitable for structured data like databases. In contrast, object storage (MinIO) does not expose files directly through the file system and instead provides access via APIs or dashboards, making it ideal for unstructured data like images.





SS5 (experiment 2)





Observation:

After modifying the code to include metadata during image upload, the uploaded image in MinIO displayed additional fields such as product name and price in the dashboard without opening the file. However, similar metadata was not visible for the database file stored in block storage, where only structured table data could be accessed through queries.

Conclusion:

Object storage supports rich, customizable metadata that can be associated with files and accessed independently of the file content. Block storage, on the other hand, only stores raw data and relies on external systems like databases to interpret and manage that data, lacking inherent metadata capabilities.

SS6 (experiment 3)

Observation:
Even after increasing storage usage, the system continued to function correctly, indicating that the storage limit had not yet been reached. However, block storage is inherently capacity-bound and depends on the fixed size of the allocated volume. When this limit is reached, database operations may fail with disk-related errors.

Concept:
Object storage systems like MinIO are designed to scale horizontally and can continue to store data independently as long as underlying resources are available. In contrast, block storage has fixed capacity constraints and does not scale as easily.

```
apharna@apharna-ASUS:~/PES2UG23CS086/my_block_data$ sudo fallocate -l 100M dummy1.bin
sudo fallocate -l 100M dummy2.bin
sudo fallocate -l 100M dummy3.bin
[sudo] password for apharna:
apharna@apharna-ASUS:~/PES2UG23CS086/my_block_data$ cd
apharna@apharna-ASUS:~$ cd Downloads
apharna@apharna-ASUS:~/Downloads$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@codmav.png" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
apharna@apharna-ASUS:~/Downloads$ cd ~/PES2UG23CS086/my_block_data
apharna@apharna-ASUS:~/PES2UG23CS086/my_block_data$ sudo fallocate -l 500M dummy4.bin
sudo fallocate -l 500M dummy5.bin
apharna@apharna-ASUS:~/PES2UG23CS086/my_block_data$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@codmav.png" http://localhost:5000/product
curl: (26) Failed to open/read local data from file/application
apharna@apharna-ASUS:~/PES2UG23CS086/my_block_data$ cd
apharna@apharna-ASUS:~$ cd Downloads
apharna@apharna-ASUS:~/Downloads$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@codmav.png" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
```

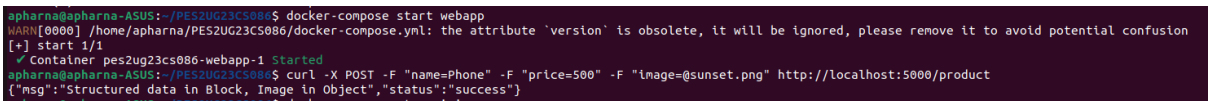
SS7 (experiment 4)

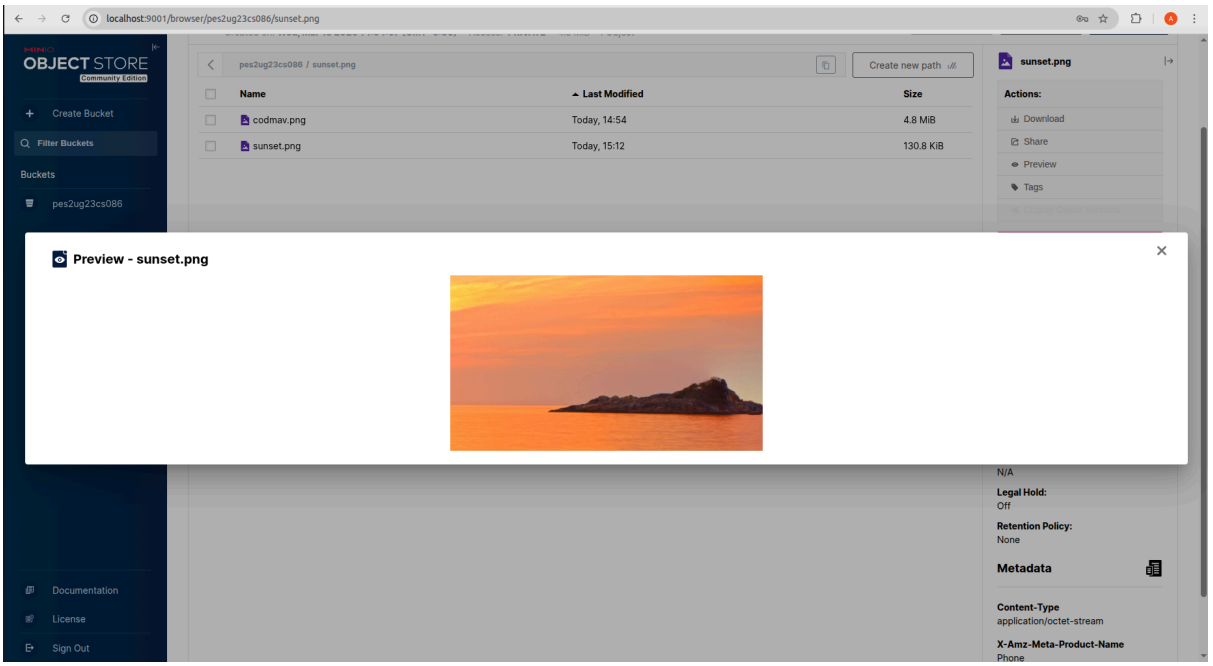
Once miniIO and webapp stopped,

```
apharna@apharna-ASUS:~/PES2UG23CS086$ sudo docker-compose stop minio
WARN[0000] /home/apharna/PES2UG23CS086/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] stop 1/1
 ✔ Container pes2ug23cs086-minio-1 Stopped
```

```
apharna@apharna-ASUS:~/Downloads$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@codmav.png" http://localhost:5000/product
<!doctype html>
<html lang=en>
<title>500 Internal Server Error</title>
<h1>Internal Server Error</h1>
<p>The server encountered an internal error and was unable to complete your request. Either the server is overloaded or there is an error in the application.</p>
apharna@apharna-ASUS:~/Downloads$
```

After restarting it,





Observation:

When the web application container was stopped, API requests failed with connection errors, indicating that the application layer was unavailable. However, the data stored in both block storage (`ecommerce.db`) and object storage (MinIO bucket) remained intact. After restarting the web application, the system resumed normal operation without any data loss.

Conclusion:

This experiment demonstrates the separation of concerns between application logic and storage. Data persistence is maintained independently of application availability, ensuring reliability and fault tolerance. Even if the application fails, the underlying storage systems continue to retain data, enabling recovery without loss.